

*Proof.* **1. Preliminaries and notation.** Set

$$\lambda := \frac{1}{p} \in (0, 1), \quad 1 - \lambda = \frac{1}{q}.$$

Hence  $p = \frac{1}{\lambda}$  and  $q = \frac{1}{1-\lambda}$ ; the relation  $\frac{1}{p} + \frac{1}{q} = 1$  is equivalent to  $q = \frac{p}{p-1}$  (or  $p = \frac{q}{q-1}$ ).

**2. Degenerate cases.** If  $a = 0$  or  $b = 0$  then  $ab = 0$  while the right-hand side of

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \tag{1}$$

is a sum of non-negative terms, so (1) holds. Equality forces both terms on the right to vanish, i.e.  $a = b = 0$ .

**3. Convex function and its conjugate.** Consider

$$\varphi : [0, \infty) \rightarrow [0, \infty), \quad \varphi(t) = \frac{t^p}{p}.$$

*Convexity.*

$$\varphi'(t) = t^{p-1}, \quad \varphi''(t) = (p-1)t^{p-2} \geq 0 \quad (p > 1, t \geq 0),$$

so  $\varphi$  is convex.

*Legendre (convex) conjugate.*

$$\varphi^*(s) = \sup_{t \geq 0} \{ st - \varphi(t) \}, \quad s \geq 0.$$

To locate the supremum differentiate the bracketed expression:

$$\frac{d}{dt} \left( st - \frac{t^p}{p} \right) = s - t^{p-1}.$$

The unique maximiser satisfies  $s = t^{p-1}$ , i.e.

$$t = s^{q-1} \quad \left( q - 1 = \frac{1}{p-1} \right).$$

Substituting this  $t$  gives

$$\begin{aligned} \varphi^*(s) &= s \cdot s^{q-1} - \frac{(s^{q-1})^p}{p} \\ &= s^q - \frac{s^q}{p} = \frac{s^q}{q}, \end{aligned}$$

where we used  $\frac{1}{p} + \frac{1}{q} = 1$ .

**4. Fenchel–Young inequality.** For any convex  $\varphi$  and  $s, t \geq 0$  one has

$$st \leq \varphi(t) + \varphi^*(s). \tag{2}$$

(See Rockafellar, \*Convex Analysis\*, Theorem23.5.)

Applying the concrete  $\varphi$  and  $\varphi^*$  from step3 to (2) yields

$$st \leq \frac{t^p}{p} + \frac{s^q}{q}. \quad (3)$$

**5. Application to the present variables.** Choose  $t = a$  and  $s = b$  in (3). Then (3) becomes exactly (1):

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

**6. Equality condition.** Equality in (2) holds iff the maximiser satisfies  $s \in \partial\varphi(t)$ . Since  $\varphi$  is differentiable on  $(0, \infty)$ ,

$$s = \varphi'(t) = t^{p-1}.$$

With  $s = b$  and  $t = a$  this reads  $b = a^{p-1}$ , i.e.

$$a^p = b^q. \quad (4)$$

Together with the degenerate case of step2 we obtain the complete equality description:

$$a = b = 0 \quad \text{or} \quad a, b > 0 \text{ and } a^p = b^q.$$

**7. Alternative elementary proof (weighted AM–GM).** Write

$$ab = (a^p)^{1/p} (b^q)^{1/q}.$$

Since  $\frac{1}{p} + \frac{1}{q} = 1$ , the weighted AM–GM inequality with weights  $\frac{1}{p}$  and  $\frac{1}{q}$  gives

$$(a^p)^{1/p} (b^q)^{1/q} \leq \frac{1}{p} a^p + \frac{1}{q} b^q,$$

which is precisely (1). Equality in weighted AM–GM occurs exactly when the two weighted terms are equal, i.e. when  $(a^p)^{1/p} = (b^q)^{1/q}$ , again equivalent to (4).

Thus Young's inequality holds for all  $a, b \geq 0$  with conjugate exponents  $p, q > 1$ , and equality occurs precisely in the cases described above.  $\square$   $\square$